THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · EASY LEVEL CHAPTER TEST SET 3

Chapter 8: **Introduction to Trigonometry** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The value of $\sin 90^\circ$ is: 1

(a) 2 **(b)** $\frac{1}{2}$ **(c)** 1 **(d)** 0

2. The value of $\sin^2 65^\circ + \cos^2 65^\circ$ is: 1

(a) 2 **(b)** 1 **(c)** 65 **(d)** 0

3. $\sin 15^\circ = \cos$ of: 1

(a) 75° **(b)** 165° **(c)** 15° **(d)** 30°

4. If $\sin A = \frac{8}{17}$, then $\csc A =$ 1

(a) $\frac{17}{8}$ **(b)** 2 **(c)** $\frac{8}{17}$ **(d)** $\frac{4}{9}$

5. In a right triangle with perpendicular 21, base 20 and hypotenuse 29, $\sin\theta =$ 1

(a) $\frac{29}{21}$ **(b)** $\frac{21}{20}$ **(c)** $\frac{21}{29}$ **(d)** $\frac{20}{29}$

6. In a right triangle with perpendicular 21, base 20 and hypotenuse 29, $\tan\theta =$ 1

(a) $\frac{21}{29}$ **(b)** $\frac{20}{29}$ **(c)** $\frac{21}{20}$ **(d)** $\frac{20}{21}$

7. The value of $\cos 0^\circ$ is: 1

(a) 1 **(b)** 0 **(c)** $\frac{1}{2}$ **(d)** 2

8. The value of $\sin^2 72^\circ + \cos^2 72^\circ$ is: 1

(a) 0 **(b)** 72 **(c)** 2 **(d)** 1

9. $\sin 70^\circ = \cos$ of: 1

(a) 110° **(b)** 70° **(c)** 85° **(d)** 20°

10. If $\sin A = \frac{7}{25}$, then $\csc A =$ 1

(a) $\frac{7}{25}$ **(b)** $\frac{24}{7}$ **(c)** $\frac{7}{26}$ **(d)** $\frac{25}{7}$

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Evaluate: $\sin^2 47^\circ + \cos^2 47^\circ$. 2

12. In a right triangle the perpendicular is 24 and the hypotenuse is 25. Find $\sin\theta$. 2

13. Find the value of $cos60^\circ + tan45^\circ$. 2

14. Evaluate: $\sin^2 62^\circ + \cos^2 62^\circ$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In $\triangle ABC$, right-angled at B, $AB = 20$ cm and $BC = 21$ cm. Find $\sin A$, $\cos A$ and $\tan A$. 3

16. If $\sin A = \frac{21}{29}$, find $\cos A$ and $\tan A$. 3

17. Evaluate: $\tan^2 45^\circ - \sin^2 30^\circ$. 3

OR

In $\triangle ABC$, right-angled at B, $AB = 9$ cm and $BC = 40$ cm. Find $\sin A$, $\cos A$ and $\tan A$. 3

18. If $\sin A = \frac{40}{41}$, find $\cos A$ and $\tan A$. 3

OR

Evaluate: $3\sin^2 90^\circ - \cos^2 0^\circ$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. If $\tan\theta = \frac{21}{20}$, find $\sin\theta$, $\cos\theta$ and $\sec\theta$. 5

OR

In $\triangle PQR$, right-angled at Q, $PQ = 60$ cm and $QR = 63$ cm. Find $\sin P$, $\cos P$, $\tan P$ and check that $\sin^2 P + \cos^2 P = 1$. 5

20. If $\tan\theta = \frac{40}{9}$, find $\sin\theta$, $\cos\theta$ and $\sec\theta$. 5

OR

In $\triangle PQR$, right-angled at Q, $PQ = 18$ cm and $QR = 80$ cm. Find $\sin P$, $\cos P$, $\tan P$ and check that $\sin^2 P + \cos^2 P = 1$. 5

END OF QUESTION PAPER — Total: 40 Marks

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